

LAB:- 02

Q1} what are the different protocols you observe at the following layers of the protocol stacks

- A. Application layer
- B. Transport layer
- C. network layer

Ans During the packet capture in Wireshark while accessing `http://neverssl.com` and `https://www.cornell.edu`. The following protocols were observed at different layers of the protocol stack.

Application layer:-

The following application layer protocols were observed:-

- * HTTP (Hypertext Transfer Protocol):-
It is used to access web pages and transfer data b/w the client and the server.
- * DNS (Domain Name System):-
Used to resolve the domain names into IP addresses before establishing the connection.
- * TLS (Transport Layer Security):-
Used to establish a secure encrypted connection while accessing `https://www.cornell.edu`.

Transport layer:-

The following transport layer protocols were observed:-

→ TCP (Transmission Control Protocol):-

Used to provide reliable communication for HTTP and HTTPS traffic.

→ UDP (User Datagram Protocol):-

Used mainly by DNS queries and responses because it is faster and suitable for small data transfers.

Network layer:-

The following network layer protocol was observed:-

* IPv4 (Internet Protocol version-4):-

Used to route packets b/w source and destination IP addresses over the network.

Q2} what is the total amount of data received for the following two cases?

A} when you access <http://neverssl.com/>

Total amount of data received while accessing http://neverssl.com = 4 KB

CN-LAB2.pcap

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http

No.	Time	Source	Destination	Protocol	Length	Info
216	6.771648000	10.85.20.68	92.223.116.254	HTTP	502	GET /filestreamingservice/files/dac53d9e-bb30-4aad-8d6a-229aa176847b?P1=1786901901&P2=404&P3=
289	6.918482700	10.85.20.68	92.223.116.254	HTTP	508	GET /filestreamingservice/files/dac53d9e-bb30-4aad-8d6a-229aa176847b?P1=1786901901&P2=404&P3=
3103	45.331220300	10.85.20.68	23.212.0.43	HTTP	165	GET /connecttest.txt HTTP/1.1
3112	45.387825100	23.212.0.43	10.85.20.68	HTTP	241	HTTP/1.1 200 OK (text/plain)
3519	47.268489000	10.85.20.68	23.53.240.17	HTTP	509	GET /filestreamingservice/files/dac53d9e-bb30-4aad-8d6a-229aa176847b?P1=1786901901&P2=404&P3=
3520	47.268682400	10.85.20.68	23.53.240.10	HTTP	503	GET /filestreamingservice/files/dac53d9e-bb30-4aad-8d6a-229aa176847b?P1=1786901901&P2=404&P3=
3648	47.744773700	10.85.20.68	59.92.30.17	HTTP	208	GET /connecttest.txt HTTP/1.1
3673	47.797776000	59.92.30.17	10.85.20.68	HTTP	243	HTTP/1.1 200 OK (text/plain)
6226	49.614052200	10.85.20.68	10.85.21.7	HTTP	289	GET /upnphost/udhisapi.dll?content=uuid:447c76a5-2c8f-4036-adfc-31f340eb91b8 HTTP/1.1
6268	49.634487200	10.85.21.7	10.85.20.68	HTTP/X.	858	HTTP/1.1 200 OK
6270	49.637692600	10.85.20.68	10.85.21.7	HTTP	308	GET /upnphost/udhisapi.dll?content=uuid:447c76a5-2c8f-4036-adfc-31f340eb91b8 HTTP/1.1
6313	49.654631000	10.85.21.7	10.85.20.68	HTTP/X.	858	HTTP/1.1 200 OK
6427	49.725111700	23.53.240.10	10.85.20.68	HTTP	1479	HTTP/1.1 206 Partial Content

> Frame 216: Packet, 502 bytes on wire (4016 bits), 502 bytes captured (4016 bits) on 0
 > Ethernet II, Src: AzureWaveTec_65:48:17 (f8:54:f6:65:48:17), Dst: ASIXElectron_35:2d:a0
 > Internet Protocol Version 4, Src: 10.85.20.68, Dst: 92.223.116.254
 > Transmission Control Protocol, Src Port: 65403, Dst Port: 80, Seq: 1, Ack: 1, Len: 436
 > Hypertext Transfer Protocol

0000 f8 54 3b 35 2d a0 f8 54 f6 65 48 17 08 00 45 00 ; 5 - T W H E
 0010 01 e8 59 8a 40 00 00 06 4f 00 0a 55 14 44 3c d6 Y @ . . . U D /
 0020 74 f6 ff 7b 00 50 1a d0 82 0f a2 bd 17 7d 8a 18 t { P } .
 0030 08 0b 56 f9 00 00 01 01 00 0a 13 3c 84 65 07 d5 V k a .
 0040 87 14 47 45 54 20 2f 66 69 6c 65 73 74 72 65 61 GET / f i l e s t r e a
 0050 6d 69 6e 07 73 65 72 76 69 63 65 2f 66 69 6c 65 m i n g s e r v i c e / f i l e
 0060 73 2f 64 61 63 35 33 64 69 65 20 62 62 33 30 2d s / d a c 5 3 d 9 e - b b 3 0 -
 0070 34 61 61 64 2d 38 64 36 61 2d 32 32 39 61 61 31 4 a a d - 8 d 6 a - 2 2 9 a a 1
 0080 37 36 30 34 37 62 3f 50 31 3c 31 37 38 36 39 30 7 5 8 4 7 b / P 1 = 1 7 8 6 9 0
 0090 31 39 30 31 26 50 12 3d 34 30 34 26 50 33 3d 32 1 9 0 1 & P 2 = 4 0 4 & P 3 = 2
 00a0 26 50 34 3d 4d 65 5a 30 47 52 60 60 40 6c 72 42 & P 4 = M e Z 0 G R n j N l r B
 00b0 4a 73 37 64 36 4f 76 38 33 77 4b 43 25 12 62 67 3 s 7 d 6 2 v 8 3 a K C 3 2 b g
 00c0 6e 50 46 37 39 25 32 66 08 67 58 4e 25 32 66 65 n P f 7 9 6 2 f h g X 3 2 f e
 00d0 72 67 33 63 64 62 38 78 47 30 46 42 57 6a 6f 25 r g 3 d o B x 3 Q F R x j o %
 00e0 32 62 52 53 73 74 63 42 62 36 79 4e 69 37 64 4b Z o R 5 e t c B b o y N i 7 d k
 00f0 5a 72 45 70 69 56 25 32 62 34 40 75 4a 54 54 4f Z r E p i v X 2 b 4 k u J T T 2
 0100 4d 39 55 41 25 33 64 25 33 64 20 48 54 54 50 2f M 9 u A 3 3 d % 3 d H T T P /
 0110 31 2a 31 8d 0a 43 6f 6a 6a 65 63 74 69 6f 6a 3a 1 . 1 : - C o n n e c t i o n :

Packets: 116226 - Displayed: 159 (0.1%) - Dropped: 0 (0.0%) - Profile Data

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tcp

No.	Time	Source	Destination	Protocol	Length	Info
181	5.443425388	10.85.20.68	23.206.178.49	TCP	54	65390 → 443 [RST, ACK] Seq=1 Ack=1 Min=0 Len=0
204	6.296787388	10.85.20.68	23.33.240.9	TCP	66	65480 → 443 [SYN] Seq=0 Min=65535 Len=0 MSS=1460 WS=256 SACK_PERM
207	6.516666388	192.178.158.188	10.85.20.68	TCP	60	5220 → 57371 [RST] Seq=1 Min=0 Len=0
213	6.639034188	10.85.20.68	106.51.156.137	TCP	66	65480 → 80 [SYN] Seq=0 Min=65535 Len=0 MSS=1460 WS=256 SACK_PERM
214	6.771231388	92.223.116.254	10.85.20.68	TCP	74	80 → 65483 [SYN, ACK] Seq=0 Ack=1 Min=31855 Len=0 MSS=1460 SACK_PERM TSval=3884257844 TSecr=
215	6.771369888	10.85.20.68	92.223.116.254	TCP	66	65483 → 80 [ACK] Seq=1 Ack=1 Min=2816 Len=0 TSval=356287589 TSecr=3884257844
216	6.771648888	10.85.20.68	92.223.116.254	HTTP	502	GET /filestreamingservice/Files/dac53d9e-bb38-4aad-8d8a-229aa176847b?P1=17869019052P2=4042P3=
217	6.780168288	57.155.104.224	10.85.20.68	TCP	68	443 → 65480 [SYN, ACK] Seq=0 Ack=1 Min=65535 Len=0 MSS=1460 WS=256 SACK_PERM
218	6.780291988	10.85.20.68	57.155.104.224	TCP	54	65482 → 443 [ACK] Seq=1 Ack=1 Min=65280 Len=0
219	6.781622288	10.85.20.68	57.155.104.224	TLSv1.2	268	Client Hello (SHA1-settings-win.data.microsoft.com)
223	6.809816688	10.85.20.68	8.8.8.8	TCP	66	63189 → 443 [SYN] Seq=0 Min=65535 Len=0 MSS=1460 WS=256 SACK_PERM
224	6.810316188	10.85.20.68	8.8.8.8	TCP	66	51183 → 443 [SYN] Seq=0 Min=65535 Len=0 MSS=1460 WS=256 SACK_PERM
225	6.839413188	10.85.20.68	22.39.35.61	TCP	66	(TCP Retransmission) 65485 → 443 [SYN] Seq=0 Min=65535 Len=0 MSS=1460 WS=256 SACK_PERM

Frame 216: Packet, 502 bytes on wire (4016 bits), 502 bytes captured (4016 bits)

- Ethernet II, Src: AzureWaveTec_65:48:17 (F8:54:F6:65:48:17), Dst: ASIXElectron_35:2d:a
- Internet Protocol Version 4, Src: 10.85.20.68, Dst: 92.223.116.254
- Transmission Control Protocol, Src Port: 65483, Dst Port: 80, Seq: 1, Ack: 1, Len: 436
- Hypertext Transfer Protocol

```

0000  f8 54 3b 35 2d a0 f8 54  f6 65 48 17 08 00 45 00  j5- T mH E
0010  01 08 59 8a 40 00 00 06  af 00 0a 55 14 44 3c d6  Y & - - U DV
0020  74 fa ff 7b 80 50 1a 0d  82 0f 42 bd 17 7d 80 18  t { P - - - }
0030  08 0b 56 f9 00 00 01 01  08 0a 13 3c 84 65 07 d6  V - - - < o
0040  87 14 47 45 54 28 2f 66  69 6c 65 73 74 72 65 61  GET /f il1stra
0050  6d 69 6e 67 73 65 72 76  69 63 63 2f 66 69 6c 65  mingserv ice/file
0060  73 2f 64 61 63 35 33 64  39 65 2d 62 62 33 30 2d  s/dac53d 9e-bb38-
0070  34 61 61 64 2d 38 64 36  61 2d 32 32 39 61 61 31  4aad-8d8 a-229aa1
0080  37 36 38 34 37 62 3f 50  31 3c 31 37 38 36 39 30  76847b?P 1=178690
0090  31 39 38 31 26 50 32 36  34 38 34 26 50 33 3d 32  1981&P2= 404&P3=2
00a0  26 50 34 3d 4d 65 5a 30  47 52 6e 6a 4e 6c 72 42  &P4=6a28 GRnjHlR
00b0  4a 73 37 64 36 4f 76 38  33 77 4b 43 25 32 62 67  3x7d6dv8 3xK%2by
00c0  6e 50 46 37 39 25 32 66  68 67 58 4a 25 32 66 65  nP779%2f hgX%2fe
00d0  72 67 33 63 64 62 38 78  47 30 46 43 57 6a 6f 25  rg3cd8x G0F8xja5
00e0  32 62 52 53 73 74 63 42  62 36 79 4e 69 37 64 4b  2bR5sc8b b6yN17d
00f0  5a 72 45 70 69 56 25 32  62 34 4b 75 4a 54 34 4f  2rEplv%2 b4KuJTT0
0100  4d 39 55 41 25 31 64 25  33 64 30 48 54 54 50 2f  M9uGX3d% 3d HTTP/
0110  31 2a 31 8d 8a 43 6f 66  6e 65 63 74 69 6f 6a 3a  1.1- Con naction:
  
```

Transmission Control Protocol Protocol

Packets: 116226 - Displayed: 75309 (65.3%) - Dropped: 0 (0.0%) Profile: Default

CN-AB2 pcap

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dns

No.	Time	Source	Destination	Protocol	Length	Info
197	6.151453500	10.85.20.68	8.8.8.8	DNS	74	Standard query 0x5dc2 A www.google.com
198	6.180617500	10.85.20.68	1.1.1.1	DNS	76	Standard query 0x132e HTTPS web.whatsapp.com
199	6.181283000	10.85.20.68	1.1.1.1	DNS	76	Standard query 0x188f A web.whatsapp.com
200	6.181649000	10.85.20.68	1.1.1.1	DNS	89	Standard query 0x8e6b HTTPS_51221_https.web.whatsapp.com
201	6.182789700	10.85.20.68	1.1.1.1	DNS	76	Standard query 0x78e5 A web.whatsapp.com
203	6.294795200	8.8.8.8	10.85.20.68	DNS	246	Standard query response 0x4cbd A assets.msn.com CNAME assets.msn.com-world-atm-default-traffic
212	6.637441100	1.1.1.1	10.85.20.68	DNS	193	Standard query response 0xb2fd A www.msftconnecttest.com CNAME www.msftconnecttest.com.edgesuite.net
220	6.791928100	8.8.8.8	10.85.20.68	DNS	158	Standard query response 0x5bdf HTTPS chrome.cloudflare-dns.com HTTPS
221	6.795779200	8.8.8.8	10.85.20.68	DNS	102	Standard query response 0x81a9 A dns.google A 8.8.8.8 A 8.8.4.4
222	6.800566700	8.8.8.8	10.85.20.68	DNS	146	Standard query response 0x9469 HTTPS dns.google SOA ns1.zdns.google
227	6.812741300	8.8.8.8	10.85.20.68	DNS	102	Standard query response 0x928a A dns.google A 8.8.8.8 A 8.8.4.4
230	6.822512200	8.8.8.8	10.85.20.68	DNS	158	Standard query response 0xd59b HTTPS chrome.cloudflare-dns.com HTTPS
234	6.842112200	8.8.8.8	10.85.20.68	DNS	146	Standard query response 0x1101 HTTPS dns.google SOA ns1.zdns.google

Frame 212: Packet, 193 bytes on wire (1544 bits), 193 bytes captured (1544 bits)

- Ethernet II, Src: ASIXElectron_35:2d:a8 (f8:e4:9b:35:2d:a8), Dst: AzureWaveTec_65:48:14 (08:00:27:65:48:14)
- Internet Protocol Version 4, Src: 1.1.1.1, Dst: 10.85.20.68
- User Datagram Protocol, Src Port: 53, Dst Port: 57868
- Domain Name System (response)

```

0000  f8 54 f6 65 48 17 f8 e4 3b 35 2d a8 00 00 45 00  T  W  . 5  .  E
0010  00 03 59 14 40 00 37 11 c9 80 01 01 01 01 0a 55  Y @ 7  . . . . . U
0020  14 44 08 35 e2 0c 00 9f f7 1c 02 fd 81 80 00 01  D 5  . . . . .
0030  00 04 00 00 00 00 03 77 77 77 0f 6d 73 66 74 63  . . . . . www.msftc
0040  6f 6a 6a 65 63 74 74 65 73 74 63 63 6f 6d 00 00  onnectte st co.
0050  01 08 01 c8 0c 00 05 00 01 00 00 0e 00 00 27 03  . . . . .
0060  77 77 77 0f 6d 73 66 74 63 6f 6a 6a 65 63 74 74  www.msft connectt
0070  65 73 74 03 63 6f 6d 09 65 64 67 65 73 75 69 74  est con edgessuit
0080  65 03 6a 65 74 00 c0 35 00 05 00 01 00 00 54 5d  a.net 5  . . . . . T)
0090  00 0f 03 61 33 39 01 64 06 63 6b 61 6d 61 69 c0  www.msft d nnecti
00a0  57 c0 60 00 01 00 01 00 00 00 11 00 04 5a 33 9c  W h . . . . . 33
00b0  89 c0 68 00 01 00 01 00 00 00 11 00 04 5a 33 9c  h . . . . . 33
00c0  93
  
```

Domain Name System Protocol

Packets: 116276 - Displayed: 352 (0.3%) - Dropped: 0 (0.0%) - Profile Data

30°C

22:07

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No.	Time	Source	Destination	Protocol	Length	Info
11	0.11852200	57.144.143.32	10.85.20.68	TLSv1.2	117	Application Data
52	0.859264700	10.85.20.68	142.251.151.119	QUIC	1292	Initial, DCID=afe574c0f2501c2, PKN: 2, PADDING, CRYPTO, CRYPTO, CRYPTO, PING, PING, CRYPTO
97	3.902670200	10.85.20.68	20.184.175.0	TLSv1.2	613	Client Hello (SNI=mobile.events.data.microsoft.com)
219	6.701622200	10.85.20.68	57.155.104.224	TLSv1.2	268	Client Hello (SNI=settings-win.data.microsoft.com)
225	6.812414900	10.85.20.68	8.8.8.8	QUIC	1292	Initial, DCID=2fbf45f1590ee256, PKN: 1, PING, PING, PING, PADDING, PING, PING, PING, CRYPTO,
226	6.812528200	10.85.20.68	8.8.8.8	QUIC	1292	Initial, DCID=2fbf45f1590ee256, PKN: 2, PADDING, PING, PING, CRYPTO, PING, CRYPTO, CRYPTO, CRYPTO,
241	6.856247600	10.85.20.68	8.8.8.8	QUIC	1292	Initial, DCID=1bec9b1646ff4a9d, PKN: 2, CRYPTO, CRYPTO, CRYPTO, PING, CRYPTO
248	6.864403100	10.85.20.68	8.8.8.8	QUIC	1292	Initial, DCID=74a6f22bf0476360, PKN: 1, PING, CRYPTO, CRYPTO
249	6.864409000	10.85.20.68	8.8.8.8	QUIC	1292	Initial, DCID=74a6f22bf0476360, PKN: 2, CRYPTO
260	6.880766000	10.85.20.68	8.8.8.8	QUIC	1292	Initial, DCID=f2449cd96c85c9ac, PKN: 2, PING, PADDING, CRYPTO, CRYPTO, PADDING, PING, PING, CRYPTO,
267	6.895533100	10.85.20.68	162.159.61.3	QUIC	1292	Initial, DCID=8e18872eb4b789c3, PKN: 2, PING, PING, PADDING, PING, PADDING, PING, PADDING, CRYPTO,
274	6.906543600	10.85.20.68	8.8.8.8	QUIC	1292	Initial, DCID=76810fffa9706d1, PKN: 2, PADDING, CRYPTO, PING, CRYPTO, PADDING, PING, CRYPTO,
282	6.916647200	10.85.20.68	172.64.41.3	QUIC	1292	Initial, DCID=438ee5b86c2157ca, PKN: 2, CRYPTO, CRYPTO, CRYPTO, CRYPTO, PADDING, PING, CRYPTO,

Frame 97: Packet, 613 bytes on wire (4904 bits), 613 bytes captured (4904 bits) on interface 0

Ethernet II, Src: AzureWaveTec_65:48:17 (f8:54:f6:55:48:17), Dst: ASIXElectron_35:2d:a0:00 (08:00:27:35:2d:a0:00)

Internet Protocol Version 4, Src: 10.85.20.68, Dst: 20.184.175.0

Transmission Control Protocol, Src Port: 65401, Dst Port: 443, Seq: 1, Ack: 1, Len: 55

Transport Layer Security

```

0000  f8 54 3b 35 2d a0 00 00  f6 65 48 17 08 00 45 00  ;5- T mH E
0010  02 57 38 67 40 80 80 06  05 08 6a 55 14 44 14 b8  U8g...U D
0020  af 08 ff 79 81 bb 01 1f  0a b1 79 a0 b4 58 30 18  y...y XP
0030  04 00 17 a6 00 00 16 03  01 01 1a 01 00 02 25 03  ...*...&
0040  83 a7 08 41 60 b6 7c 46  3c 60 70 40 90 45 a8 9d  A |F=p.F
0050  74 87 76 b3 1e 44 21 bb  cc 04 3c 36 95 29 89 d5  t v D! ...6 }
0060  62 20 33 84 95 49 64 57  86 07 91 08 58 07 a7 2f  b 3 Idn ...X /
0070  eb e2 97 cb eb 9b c7 9c  54 3e ca 28 02 75 09 e2  T < u
0080  1d a9 00 28 13 02 13 01  c0 2c c0 2b c0 30 c0 2f  { ... + B /
0090  c0 24 c0 23 c0 28 c0 27  c0 0a c0 09 c0 14 c0 13  $ # { '
00a0  00 3d 00 9c 00 3d 00 3c  00 35 00 2f 01 00 01 05  ...< 5 /
00b0  00 00 00 25 00 23 00 00  20 60 6f 62 69 61 65 2e  % # mobilw
00c0  65 76 65 6e 74 73 2e 64  01 74 62 2e 60 69 63 72  events.d ata,mic
00d0  6f 73 6f 66 74 24 63 6f  60 00 05 00 05 01 00 00  osoft.ca m
00e0  00 00 00 2b 00 05 04 03  04 01 03 00 0d 00 1a 00  A
00f0  16 88 04 08 05 06 06 64  01 05 01 02 01 04 03 05  ... ..
0100  83 82 83 82 82 86 81 86  03 00 23 00 00 00 0a 00  H
0110  08 08 08 08 1d 06 17 60  18 00 00 00 02 01 00 00  ... ..
  
```

Packets: 116/226 · Displayed: 863 (0.75%) · Dropped: 0 (0.0%) · Filter: Data

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No.	Time	Source	Destination	Protocol	Length	Info
83	3.788273200	8.8.8.8	10.85.20.68	DNS	205	Standard query response 0x7632 A us-west-2.kinesis.hpanalytics.net CHAVE d-7anea20jyd.execute
87	3.883156800	10.85.20.68	51.89.13.61	TCP	66	65403 → 443 [SYN] Seq=0 Min=65535 Len=0 MSS=1460 WS=256 SACK_PERM
93	3.891890100	10.85.20.68	20.184.175.0	TCP	66	[TCP Retransmission] 65402 → 443 [SYN] Seq=0 Min=65535 Len=0 MSS=1460 WS=256 SACK_PERM
94	3.922967300	10.85.20.68	57.155.104.224	TCP	66	65402 → 443 [SYN] Seq=0 Min=65535 Len=0 MSS=1460 WS=256 SACK_PERM
95	3.978817800	20.184.175.0	10.85.20.68	TCP	65	443 → 65401 [SYN, ACK] Seq=0 Ack=1 Min=65535 Len=0 MSS=1440 WS=256 SACK_PERM
96	3.978247900	10.85.20.68	20.184.175.0	TCP	54	65401 → 443 [ACK] Seq=1 Ack=1 Min=262144 Len=0
97	3.982678200	10.85.20.68	20.184.175.0	TLSv1.2	613	Client Hello [SHA256, events.data.microsoft.com]
98	3.986638600	10.85.20.68	92.223.116.254	TCP	74	65403 → 80 [SYN] Seq=0 Min=65535 Len=0 MSS=1460 WS=256 SACK_PERM TSval=3356284805 TSecr=0
99	4.002809000	10.85.20.68	92.223.116.254	TCP	74	65404 → 80 [SYN] Seq=0 Min=65535 Len=0 MSS=1460 WS=256 SACK_PERM TSval=3356284823 TSecr=0
100	4.144806100	10.85.20.68	8.8.8.8	DNS	85	Standard query 0x75d0 HTTPS chrome.cloudflare-dns.com
101	4.145151500	10.85.20.68	8.8.8.8	DNS	85	Standard query 0xba2e A chrome.cloudflare-dns.com
102	4.145508800	10.85.20.68	8.8.8.8	DNS	70	Standard query 0x9469 HTTPS dns.google
103	4.145814300	10.85.20.68	8.8.8.8	DNS	70	Standard query 0x81a9 A dns.google

Frame 97: Packet, 613 bytes on wire (4904 bits), 613 bytes captured (4904 bits) on interface 0

Ethernet II, Src: AzureNaveTec_65:48:17 (f8:54:f6:65:48:17), Dst: ASIXElectron_35:2d:a:0 (08:00:27:35:2d:a0)

Internet Protocol Version 4, Src: 10.85.20.68, Dst: 20.184.175.0

Transmission Control Protocol, Src Port: 65401, Dst Port: 443, Seq: 1, Ack: 1, Len: 55

Transport Layer Security

0000 f8 54 3b 35 2d a0 f8 54 f6 65 48 17 08 00 45 00 ;5- T m E
 0010 02 57 38 67 40 00 00 06 65 48 0a 55 14 44 14 b8 W000 ... U D
 0020 a7 00 ff 79 01 bb 01 1f 0a b2 79 ad b4 58 50 18 y ... y XP
 0030 04 08 27 c6 00 00 16 03 01 02 1a 01 00 02 26 05 ... + S
 0040 03 a7 a8 41 60 b6 7c 46 3c 60 70 ab 90 46 a8 9d A F = p F
 0050 74 07 76 b3 1e 44 21 bb cc 04 5c 36 95 29 89 d6 t v D! ... G)
 0060 62 20 33 84 95 49 64 57 86 07 91 08 58 07 07 2f b 3 IdA ... X - /
 0070 eb e2 97 cb eb 9b c7 9c 54 1e ca 28 02 75 09 e2 T> (u)
 0080 1d a9 00 28 13 02 13 01 c0 2c c0 2b c0 30 c0 2f { ... + 0 /
 0090 c0 24 c0 23 c0 28 c0 27 c0 0a c0 09 c0 14 c0 13 \$ * (' 5 / ...
 00a0 00 9d 00 9c 00 3d 00 3c 00 33 00 2f 01 00 01 b5 ... = < 5 / ...
 00b0 00 00 00 25 00 23 00 00 20 6d 6f 62 69 61 65 2e % # ... oblle:
 00c0 65 76 65 66 74 73 2e 64 01 74 61 2e 6d 69 63 72 events.d ata.ricr
 00d0 6f 73 ef 66 74 2e 63 6f 6d 00 05 00 05 01 00 00 osoft.co m
 00e0 00 00 00 2b 00 05 04 03 04 01 05 00 00 00 1a 00 +
 00f0 18 00 04 00 05 00 06 64 01 05 01 01 04 05 05
 0100 03 02 03 02 02 06 01 06 01 00 73 00 00 00 0a 00 #
 0110 00 00 00 00 1d 00 17 00 18 00 00 00 02 01 00 00

Internet Protocol Version 4: Protocol

Packets: 116276 - Displayed: 88222 (75.9%) - Dropped: 0 (0.0%) - Profile: Default

Windows taskbar showing search bar, task icons, system tray with date 15-08-2025 and time 22:08.

The following network layer protocol was observed:-

* IPV4 (Internet Protocol version-4):-

Used to route packets b/w source and destination IP addresses over the network.

Q2} what is the total amount of data received for the following two cases?

A} when you access <http://neverssl.com/>

Total amount of data received while accessing http://neverssl.com = 4 KB

cap

Ethernet - 1

IPv4 - 1

IPv6

TCP - 1

UDP

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start
10.85.20.68	55147	34.223.124.45	80	3	4 kB	155	16	18.75%	0	0 bytes	3	4 kB	481.610237

2556453380

1461 (relative sequence number)]

77 (relative ack number)

0000 33 3a 33 33 20 47 4d 54 0d 0a 45 54 61 67 3a 20 3:33 GMT ETag:

00e0 22 66 37 39 2d 35 65 32 38 62 32 39 64 33 38 65 "f79-5e2 8b29d38e

00f8 39 33 22 0d 0a 41 63 63 65 70 74 2d 52 61 6e 67 93" Acc ept-Rang

0100 65 73 3a 20 62 79 74 65 73 0d 0a 43 6f 6e 74 65 es: byte s Conte

0110 6e 74 2d 4c 65 6e 67 74 68 3a 20 33 39 36 31 0d nt-Lengt h: 3961

Packets: 116226

Displayed: 3 (0.0%)

Dropped: 0 (0.0%)

Profile: Default

B. when you access ~~http~~ $\text{https}://\text{www.cornell.edu}$

Ans:- The total amount of data received with $\text{https}://\text{www.cornell.edu}$ =

* Byte $A \rightarrow B = 0 \text{ KB}$

* Byte $B \rightarrow A = 4 \text{ KB}$

Q 3} Answer the following when you access the site `http://neverxl.com/`.

A} How many HTTP GET requests do you observe? List down the GET requests.

Ans:- Total HTTP GET Requests = 1

B} what version of HTTP the server is running?

HTTP version = HTTP/1.1

C} For each of the HTTP GET requests, find out:-

i} The total number of TCP segments being received.

Ans 3 TCP segments were received from the server.

ii} Total amount of data received in the corresponding HTTP response:-

Total data received = 4261 bytes

Q. Identify TCP three way handshake used to establish the connection with the <http://never081.com>. Specify SYN, SYN-ACK, ACK packets and write the sequence number for each of the three accesses.

Ans.

<u>Packet</u>	<u>Flags</u>	<u>Seq. no.</u>	<u>Port</u>
Packet 3196	SYN	0	
Packet 3470	SYN, ACK	0	
Client ACK	ACK	1	

Current filter: ip.dst == 34.223.124.45 && http.request.method == "GET"

No.	Time	Source	Destination	Protocol	Length	Info
111983	497.084671500	10.85.20.68	34.223.124.45	HTTP	130	GET / HTTP/1.1

> Frame 111983: Packet, 130 bytes on wire (1040 bits), 130 bytes captured (1040 bits)

> Ethernet II, Src: AzureWaveTec_65:48:17 (f8:54:f6:65:48:17), Dst: ASIXElectron_35:2d:7c (08:00:27:00:00:2f)

> Internet Protocol Version 4, Src: 10.85.20.68, Dst: 34.223.124.45

> Transmission Control Protocol, Src Port: 55147, Dst Port: 80, Seq: 1, Ack: 1, Len: 76

Source Port: 55147
Destination Port: 80
[Stream index: 155]
[Stream Packet Number: 8]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 76]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 791262763
[Next Sequence Number: 77 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)

0000 f8 e4 3b 35 2d a0 f8 54 f6 65 48 17 08 00 45 00 ;S- T eH E
0010 00 74 40 d3 40 00 80 06 fc 0b 0a 55 14 44 22 df t@ U.D
0020 7c 2d d7 6b 00 50 2f 29 b6 2b 98 60 62 04 50 18 | - k P /) + ' b P
0030 ff ff bf e5 00 00 47 45 54 20 2f 20 48 54 54 50 GE T / HTTP
0040 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 6e 65 76 65 /1.1- Ho st: neve
0050 72 73 73 6c 2e 63 6f 6d 0d 0a 55 73 65 72 2d 41 rssl.com -User-A
0060 67 65 6e 74 3a 20 63 75 72 6c 2f 38 2a 32 31 2e gent: cu r1/8.21.
0070 30 0d 0a 41 63 63 65 70 74 3a 20 2a 2f 2a 0d 0a 0 -Accep t: */*
0080 0d 0a

Packets: 116226 - Displayed: 1 (0.0%) - Dropped: 0 (0.0%) Profile: Default